WORKED SOLUTION

STEP I 2000, Question 8

Pure • Integrals of Moduli

STEP I 2000, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s1-q8

The key idea

A modulus sign inside an integral is dealt with by finding where the expression inside changes sign. On each interval between sign changes, $|f|$ is either f or $-f$, so we split the range there and integrate ordinary functions. All the real work is in locating the zeros: by inspection for xe^x, by factorising for the cubic, and by writing $\sin x + \cos x$ as a single sine wave.

Solution

The first integral

Since $e^x > 0$ for every x, the sign of xe^x is the sign of x. So on $[-1, 0]$ we have $xe^x \leq 0$ and $|xe^x| = -xe^x$, while on $[0, 1]$ we have $xe^x \geq 0$ and $|xe^x| = xe^x$. Splitting the range at 0,

$$\int_{-1}^{1} |xe^x| \, dx = \int_{-1}^{0} (-xe^x) \, dx + \int_{0}^{1} xe^x \, dx = -\int_{-1}^{0} xe^x \, dx + \int_{0}^{1} xe^x \, dx,$$

which is the required identity.

To evaluate it, integrate by parts with $u = x$ and $dv/dx = e^x$:

$$\int xe^x \, dx = xe^x - \int e^x \, dx = (x-1)e^x + c.$$

Hence

$$\int_{-1}^{0} xe^x \, dx = [(x-1)e^x]_{-1}^{0} = -1 - (-2e^{-1}) = \frac{2}{e} - 1,$$
$$\int_{0}^{1} xe^x \, dx = [(x-1)e^x]_{0}^{1} = 0 - (-1) = 1.$$

The first of these is negative (as it must be, since the integrand is negative there), and

$$\int_{-1}^{1} |xe^x| \, dx = \left(1 - \frac{2}{e}\right) + 1 = 2 - \frac{2}{e}.$$

Part (i)

Let $p(x) = x^3 - 2x^2 - x + 2$. Grouping terms,

$$p(x) = x^2(x-2) - (x-2) = (x-2)(x^2-1) = (x+1)(x-1)(x-2).$$

The only zeros in $[0,4]$ are $x = 1$ and $x = 2$. A positive-leading cubic with simple roots $-1, 1, 2$ is positive between -1 and 1, negative between 1 and 2, and positive beyond 2. So

$$|p(x)| = \begin{cases} p(x), & 0 \le x \le 1, \\ -p(x), & 1 \le x \le 2, \\ p(x), & 2 \le x \le 4. \end{cases}$$

Take the antiderivative

$$F(x) = \tfrac{1}{4}x^4 - \tfrac{2}{3}x^3 - \tfrac{1}{2}x^2 + 2x,$$

for which

$$F(0) = 0, \qquad F(1) = \tfrac{1}{4} - \tfrac{2}{3} - \tfrac{1}{2} + 2 = \tfrac{13}{12},$$
$$F(2) = 4 - \tfrac{16}{3} - 2 + 4 = \tfrac{2}{3}, \qquad F(4) = 64 - \tfrac{128}{3} - 8 + 8 = \tfrac{64}{3}.$$

Ask yourself Why can't we just compute $F(4) - F(0)$ and take its modulus?

Because the modulus has to be applied to the integrand, not to the answer. On $[1,2]$ the cubic is negative, and $F(4) - F(0)$ would subtract that area instead of adding it. So we treat the three pieces separately:

$$\begin{aligned} \int_0^4 |p(x)|\,\mathrm{d}x &= (F(1) - F(0)) - (F(2) - F(1)) + (F(4) - F(2)) \\ &= \tfrac{13}{12} - (\tfrac{2}{3} - \tfrac{13}{12}) + (\tfrac{64}{3} - \tfrac{2}{3}) \\ &= \tfrac{13}{12} + \tfrac{5}{12} + \tfrac{62}{3} = \tfrac{3}{2} + \tfrac{62}{3} = \frac{133}{6}. \end{aligned}$$

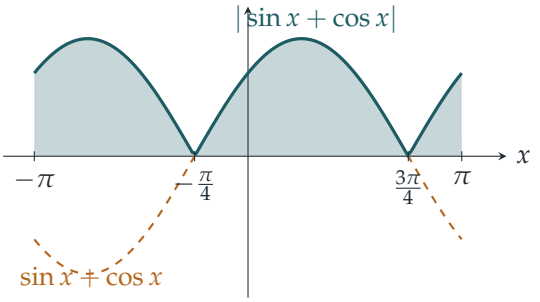
Each bracket is positive, as it should be: $\frac{13}{12}$, $\frac{5}{12}$ and $\frac{62}{3}$ are the three separate areas. (For comparison, $F(4) - F(0) = \frac{64}{3}$, which is smaller by $2 \times \frac{5}{12}$.)

Part (ii)

Using $\sin(x + \frac{\pi}{4}) = \frac{1}{\sqrt{2}}(\sin x + \cos x)$,

$$\sin x + \cos x = \sqrt{2}\,\sin(x + \tfrac{\pi}{4}).$$

This vanishes at $x = -\frac{\pi}{4}$ and $x = \frac{3\pi}{4}$ in $[-\pi, \pi]$. It is negative on $[-\pi, -\frac{\pi}{4}]$ and on $[\frac{3\pi}{4}, \pi]$, and positive in between.



We could split into three pieces as before, but there is a neater route. Substitute $u = x + \frac{\pi}{4}$:

$$\int_{-\pi}^{\pi} |\sin x + \cos x| \, \mathrm{d}x = \sqrt{2} \int_{-3\pi/4}^{5\pi/4} |\sin u| \, \mathrm{d}u.$$

The function $|\sin u|$ has period π, and the new range has length 2π, so it covers exactly two periods. The integral of a periodic function over a whole number of periods doesn't depend on where we start, so

$$\sqrt{2} \int_{-3\pi/4}^{5\pi/4} |\sin u| \, \mathrm{d}u = \sqrt{2} \times 2 \int_{0}^{\pi} \sin u \, \mathrm{d}u = \sqrt{2} \times 2 \times 2 = 4\sqrt{2}.$$

If you prefer the direct split, $G(x) = \sin x - \cos x$ is an antiderivative of $\sin x + \cos x$, with $G(-\pi) = 1$, $G(-\frac{\pi}{4}) = -\sqrt{2}$, $G(\frac{3\pi}{4}) = \sqrt{2}$ and $G(\pi) = 1$. The three areas are $1 + \sqrt{2}$, $2\sqrt{2}$ and $\sqrt{2} - 1$, which add to $4\sqrt{2}$.

Checks

- Simpson's rule on each piece (where the integrand is smooth) gives 1.264241… for the first integral, matching $2 - 2/\mathrm{e} \approx 1.264241$.
- For (i), numerical integration gives 22.16667, matching $\frac{133}{6}$. It is also slightly more than $F(4) - F(0) = \frac{64}{3} \approx 21.33$, as it must be, since the modulus turns a small negative area into a positive one.
- For (ii), numerical integration gives 5.656854, matching $4\sqrt{2}$. A sanity check: $|\sin x + \cos x|$ has the same shape as $\sqrt{2}|\sin x|$, which averages $\frac{2\sqrt{2}}{\pi}$, and $2\pi \times \frac{2\sqrt{2}}{\pi} = 4\sqrt{2}$.

Common slips

- Integrating straight across a sign change and taking the modulus at the end. This undercounts every negative region.
- Missing the root at $x = 1$ in (i). Always factorise fully, or at least check small integer roots with the factor theorem.
- Sign errors in $\left[(x-1)\mathrm{e}^x\right]_{-1}^{0}$: at $x = -1$ the value is $-2\mathrm{e}^{-1}$, so subtracting it adds $2/\mathrm{e}$.

Beyond STEP

The integral $\int_a^b |f(x)|\,\mathrm{d}x$ is the total area between a curve and the axis. At university it reappears as the L^1 norm of f, one way of measuring the size of a function and the distance between two functions. Proving that $|f|$ is integrable whenever f is, and that $\left|\int f\right| \leq \int |f|$, is a standard early exercise in real analysis. Abbott's *Understanding Analysis* develops the Riemann integral carefully enough to prove both.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.